

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Data Science and Analytics professional with **5+ years of experience** transforming complex business questions into actionable insights through statistical analysis, customer behavior analytics, experimentation, and data-driven decision-making.
- Advanced proficiency in **SQL, Python, R, Amazon Redshift, and Apache Spark**, with experience analyzing large-scale structured and semi-structured datasets containing millions of customer, transaction, product, and behavioral records.
- Strong experience designing analytical studies around **customer behavior, engagement, retention, churn, subscription trends, product adoption, cohort performance, and customer lifecycle metrics** to identify opportunities for improving user experience and business outcomes.
- Hands-on expertise in **A/B testing, hypothesis testing, statistical significance, experiment design, regression analysis, segmentation, cohort analysis, predictive modeling, and machine learning** for product and business analytics.
- Experienced in developing reusable and automated analytical frameworks using **Python, SQL, Spark, and data pipelines**, allowing recurring studies, KPIs, experiments, and customer insights to remain continuously updated as data volumes grow.
- Proven ability to work independently in **ambiguous, fast-paced environments**, proactively investigating open-ended questions, validating assumptions, iterating on analytical approaches, and communicating defensible recommendations to cross-functional stakeholders.
- Strong written and verbal communicator experienced in creating **technical documentation, analytical reports, white papers, dashboards, methodology documents, and executive-ready findings** explaining what was discovered, how results were derived, and how teams can take action.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Analytics:** Python, R, SQL, Java, Pandas, NumPy, SciPy, Scikit-learn, Jupyter Notebook, Statistical Analysis, Exploratory Data Analysis (EDA).
- **Databases & Big Data:** Amazon Redshift, PostgreSQL, MySQL, Oracle, SQL Server, Apache Spark, PySpark, Spark SQL, large-scale data processing, terabyte-scale datasets.
- **Data Science & Machine Learning:** Regression, Classification, Clustering, Predictive Modeling, Feature Engineering, Model Evaluation, Customer Segmentation, Behavioral Modeling, Machine Learning.
- **Experimentation & Statistics:** A/B Testing, Experiment Design, Hypothesis Testing, Statistical Significance, Confidence Intervals, T-Tests, Chi-Square Tests, Control/Treatment Analysis, Causal Analysis.
- **Product & Customer Analytics:** Customer Behavior Analytics, Subscription Analytics, Retention, Churn, Engagement, Cohort Analysis, Funnel Analysis, Customer Lifetime Value, Product Adoption, KPI Development.
- **Data Engineering & Automation:** ETL/ELT, Automated Analytical Pipelines, Data Transformation, Data Quality, Data Validation, AWS S3, AWS Lambda, Airflow, REST APIs, Batch Processing.
- **Visualization, Communication & Tools:** Tableau, Power BI, Matplotlib, Excel, Git, GitLab, Jenkins, Jira, Confluence, Splunk, Technical Documentation, White Papers, Agile/Scrum.

EXPERIENCE

Software Engineer

Oct 2024 – Present

PCMI LLC (Full Time)

Chicago, IL, USA

- Performed comprehensive **customer and product behavior analysis** using Python, SQL, and Amazon Redshift to evaluate engagement patterns, contract activity, service utilization, retention behavior, operational trends, and customer lifecycle metrics across large enterprise datasets.
- Translated ambiguous questions from product, operations, and business stakeholders into structured **analytical studies**, independently defining hypotheses, extracting relevant datasets, validating assumptions, performing statistical analysis, and presenting actionable recommendations.
- Developed reusable Python and SQL analytical frameworks for **cohort analysis, customer segmentation, retention analysis, trend analysis, funnel metrics, and behavioral analytics**, enabling recurring studies to automatically refresh as new customer and transactional data became available.
- Designed and analyzed **A/B tests and controlled experiments**, defining treatment and control populations, success metrics, sample criteria, hypothesis tests, confidence intervals, and statistical significance to evaluate product and workflow changes.
- Built predictive and descriptive models using **Python, Pandas, NumPy, Scikit-learn, and statistical techniques** to identify behavioral patterns, customer segments, anomaly indicators, and factors associated with customer engagement and operational outcomes.
- Leveraged **Apache Spark and PySpark** to process and aggregate high-volume datasets efficiently, preparing analysis-ready customer, transaction, contract, and application-event datasets for statistical studies and machine-learning workflows.
- Created automated data-quality and analytical validation routines using **SQL and Python** to detect missing records, duplicates, abnormal distributions, data inconsistencies, outliers, and metric deviations, ensuring analytical conclusions were accurate and reproducible.
- Authored detailed **analytical documentation, methodology reports, technical findings, and executive-ready presentations** explaining study objectives, datasets, statistical methods, assumptions, limitations, findings, and recommended business actions to technical and non-technical audiences.

Environments: Python, R, SQL, Amazon Redshift, Apache Spark, PySpark, Pandas, NumPy, SciPy, Scikit-learn, A/B Testing, Statistical Analysis, Machine Learning, Customer Analytics, Cohort Analysis, AWS S3, Jupyter Notebook, Tableau, Power BI, Git, Jira, Confluence, Agile/Scrum.

Software Engineer

Dec 2023 – Sept 2024

Client: CVS

Remote, USA

- Analyzed large-scale **member, transaction, digital interaction, and healthcare application datasets** using SQL, Python, and Redshift to understand customer behavior, engagement patterns, service utilization, retention trends, and product adoption.
- Conducted **exploratory data analysis (EDA)** using Python, Pandas, NumPy, and SQL to identify behavioral patterns, anomalies, correlations, customer segments, and emerging trends that informed product and operational decision-making.
- Designed **A/B testing methodologies and experimental studies** for digital product enhancements, establishing hypotheses, control and treatment populations, success metrics, statistical significance criteria, and post-experiment analysis.
- Developed automated analytical workflows to calculate and monitor recurring **KPIs, customer cohorts, conversion funnels, engagement metrics, retention rates, and behavioral trends**, reducing repetitive analysis and maintaining continuously updated insights.
- Applied **machine-learning and statistical modeling** techniques including regression, classification, clustering, customer segmentation, and feature engineering to identify behavioral drivers and support data-driven business decisions.
- Processed large datasets with **Apache Spark, PySpark, Spark SQL, and AWS-based data services**, performing distributed transformations and aggregations required for scalable customer and product analytics.
- Collaborated with **product managers, engineers, analysts, business stakeholders, and cross-functional teams** to convert open-ended business questions into measurable analytical problems and independently investigate potential solutions.
- Produced clear **technical documentation, analytical reports, experiment summaries, dashboards, and stakeholder presentations**, communicating methodology, assumptions, statistical evidence, findings, and recommended actions to audiences with varying technical backgrounds.

Environments: Python, R, SQL, Amazon Redshift, Apache Spark, PySpark, Pandas, NumPy, Scikit-learn, Jupyter Notebook, AWS, A/B Testing, Statistical Analysis, Customer Segmentation, Cohort Analysis, Machine Learning, Tableau, Power BI, GitLab, Jira, Confluence, Agile/Scrum.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Analyzed enterprise **product usage, customer activity, transactional, and operational datasets** using Python and SQL to identify usage patterns, anomalies, adoption trends, and opportunities for improving application and customer experiences.
- Created complex SQL queries involving **joins, window functions, CTEs, aggregations, subqueries, and statistical calculations** to retrieve, transform, and analyze high-volume data from relational and analytical databases.
- Performed **exploratory and statistical analysis** to investigate business questions, evaluate distributions and correlations, identify abnormal patterns, and validate hypotheses using Python, Pandas, NumPy, and visualization libraries.
- Built customer and product-level analytical datasets supporting **segmentation, cohort analysis, trend analysis, feature adoption, usage frequency, and operational KPI measurement** for business and product stakeholders.
- Supported **A/B and comparative analytical studies** by defining evaluation metrics, comparing user populations, measuring differences in product behavior, and validating whether observed results were statistically meaningful.
- Used **Apache Spark and PySpark** for distributed transformation and aggregation of large application datasets, enabling scalable analytical processing beyond traditional relational database workloads.
- Automated recurring data preparation, validation, and reporting workflows using **Python and SQL**, reducing manual analysis while maintaining consistency and reproducibility across recurring business studies.
- Collaborated with globally distributed **engineering, product, QA, and business teams** and documented analysis methodology, data definitions, assumptions, findings, and recommendations to support transparent and evidence-based decisions.

Environments: Python, R, SQL, Apache Spark, PySpark, Pandas, NumPy, Statistical Analysis, A/B Testing, Machine Learning, Customer Analytics, Oracle, PostgreSQL, MySQL, Jupyter Notebook, Tableau, Git, Jenkins, Jira, Confluence, Agile/Scrum.

Associate Software Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Used **SQL and Python** to extract, cleanse, transform, and analyze customer, transaction, application, and operational datasets supporting enterprise reporting and data-driven business analysis.
- Developed advanced SQL queries involving **multi-table joins, aggregations, CTEs, subqueries, window functions, and data reconciliation logic** to investigate business questions and validate customer and transaction-level information.
- Performed **exploratory data analysis** to identify patterns, anomalies, trends, and data-quality issues using Python, Pandas, NumPy, SQL, and visualization techniques, supporting investigation of application and business behavior.
- Supported development of analytical datasets used for **customer segmentation, trend analysis, operational metrics, transaction analysis, and recurring business reporting**, ensuring consistent data definitions and accurate calculations.
- Created automated Python and SQL routines for recurring **data extraction, validation, reconciliation, KPI generation, and analytical reporting**, reducing manual effort and improving repeatability of business analysis.
- Applied foundational **statistical techniques and hypothesis-driven analysis** to compare datasets, evaluate changes in application behavior, analyze distributions, and support evidence-based investigation of operational issues.
- Worked with cross-functional and geographically distributed teams to translate **business requirements and ambiguous problem statements** into data queries, measurable metrics, analytical investigations, and clearly documented findings.
- Prepared detailed documentation covering **data sources, SQL logic, analytical methodology, metric definitions, assumptions, validation steps, results, and recommendations**, enabling stakeholders to independently understand and reproduce analytical findings.

Environments: Python, SQL, R, Pandas, NumPy, Statistical Analysis, Exploratory Data Analysis, Customer Analytics, A/B Testing, Apache Spark, Oracle, MySQL, PostgreSQL, Jupyter Notebook, Tableau, Git, Jenkins, Jira, Confluence, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.